



DataWedge 3.0

Basic Configuration Guide

DataWedge Basic Configuration Guide

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Patents

This product is covered by one or more of the patents listed on the website: www.symbol.com/patents

Warranty

Revision History

Changes to the original manual are listed below:

Change	Date	Description
Rev 1	10/2008	Initial Draft

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About This Guide

Introduction

DataWedge is a Motorola mobile device application that reads data from input devices and sends it as keystrokes to consumer applications executing in the foreground on the mobile devices.

DataWedge runs on Windows CE 5.0, Windows Mobile 5 and Windows Mobile 6 based Motorola mobile devices.

DataWedge provides two views into its configuration options; a Basic view and an Advanced view. The Basic view provides more direct and more rapid access to a similar set of configurations options found on prior versions of DataWedge. The Basic view is intended as a starting point to the myriad of configuration options that are available and for those that do not need the advanced features.

This document describes the features and functionality available in the Basic view and explains how it is used to configure the various options.

A separate user guide is available which covers the Advanced view and its configuration options in detail. Refer to the *DataWedge Advanced Configuration Guide* for more information.

Notational Conventions

The following conventions are used in this document:

- “device” refers to any Motorola enterprise mobility device.
- “User” refers to anyone using an application on the device.
- “You” refers to the End User, System Administrator or Technical Support person using this manual as a reference to install, configure, operate, maintain and troubleshoot DataWedge.
- *Italics* are used to highlight the following:
 - Chapters and sections in this and related documents
 - Dialog box, window and screen names
 - Drop-down list and list box names
 - Check box and radio button names
 - Icons on a screen.

- **Bold** text is used to highlight the following:
 - Key names on a keypad
 - Button names on a screen or window.
- bullets (•) indicate:
 - Action items
 - Lists of alternatives
 - Lists of required steps that are not necessarily sequential
- Sequential lists (e.g., those that describe step-by-step procedures) appear as numbered lists.



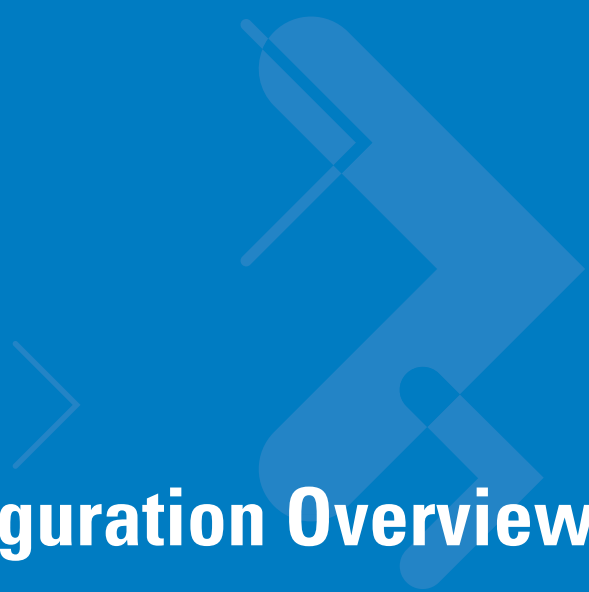
NOTE This symbol indicates something of special interest or importance to the reader. Failure to read the note will not result in physical harm to the reader, equipment or data.



CAUTION This symbol indicates that if this information is ignored, the possibility of data or material damage may occur.



WARNING! This symbol indicates that if this information is ignored the possibility that serious personal injury may occur.



Chapter 1 Basic Configuration Overview

Introduction

DataWedge provides two views into its configuration options; a Basic view and an Advanced view. Whilst the Advanced view allows access to all the configuration options, the Basic view provides more direct and more rapid access to a similar set of configurations options found on prior versions of DataWedge. The Basic view is intended as a starting point to the myriad of configuration options that are available and for those who have no need of the advanced features.

One of the new features introduced in this version of DataWedge is Profiles. Profiles allow settings to be configured per application so that when that application comes to the foreground DataWedge uses the settings specific to it as defined in the profile.

DataWedge has a default profile, Profile0, which is used when a foreground application does not have a profile associated to it. Profile0 is a generic profile that applies to all applications unless an application has its own profile.

The Basic view is a shortcut to selected configuration options within the default profile, Profile0, exposing a similar set of settings as found on previous version of DataWedge. These same settings are exposed and can be modified through the Advanced view. It is therefore important to note that changes made in one view can affect the other, especially in the case of making changes through the Advanced view to settings which are not exposed in the Basic view such as, for example, disabling the default profile.

The Basic view exposes the Barcode input plug-in, Basic Format process plug-in and Keystroke output plug-in configuration options.

Please refer to [Basic and Advanced View Considerations](#) for more details.

Profile0

Profile0 is the default profile. The Basic view allows the configuration of the default profile's barcode input plug-in, keystroke output plug-in and Basic format process plug-in as described below.

Barcode Input Plug-in

The Barcode input plug-in is responsible for reading data from the integrated barcode reader. The Barcode plug-in supports different types of barcode readers including laser, imager and camscan (Camera Scan). Raw data read from the barcode reader can be processed or formatted using process plug-ins (See [Basic Format Process Plug-in](#)) as required. DataWedge has built-in feedback functionality for the barcode reader to issue user alerts. The feedback settings can be configured according to user requirement.

Keystroke Output Plug-in

The Keystroke output plug-in is responsible for dispatching processed data to the foreground application on the mobile device by emulating keystrokes. Similar to the Barcode Input Plug-in, the Keystroke output plug-in also has feedback functionality.

Basic Format Process Plug-in

The Basic Format (aka Prefix/Suffix) plug-in is similar to the prefix/suffix feature that exists in earlier versions of DataWedge and it allows DataWedge to add either a predefined prefix or a suffix to the captured data before passing it to an output plug-in.

The Basic Format process plug-in allows setting a string, sticky keys, virtual keys, control characters (characters sent by pressing Ctrl key) and escape sequences (See [Escape Sequences Supported by DataWedge](#)) at the beginning or at the end of the data received from the input plug-in. Also this process plug-in can be used to send data in hexadecimal format, append TAB and/or ENTER keys or restrict sending data.

Basic and Advanced View Considerations

As explained above Basic view is a shortcut to selected configuration options within the default profile, Profile0, of DataWedge. The remaining settings such as Advanced Data Formatting and the enabled/disabled state are only configurable through the Advanced view.

Through the advanced view, general settings such as the enabling/disabling of profiles, Profile Selection (Auto/Manual) and Manual Profile may affect the default profile in a way that is not exposed in the Basic view.

After using the Advanced view, in order to use the Basic view successfully, please make sure the following settings are configured via the Advanced view as shown in the following table.

Table 1-1 *Basic and Advanced View Considerations*

Setting	Expected Value
Advanced > Profiles > Profile 0 > Enabled	Ticked. (Profile0 needs to be enabled and this can only be done through the Advanced view)
Advanced > Settings > Profile Selection	Manual
Advanced > Settings > Manual Profile	Profile0

Chapter 2 Getting Started

Installation

DataWedge can be installed on a mobile device via a PC or by copying the DataWedge.cab file found in C:\Program Files\Motorola DataWedge\Cab\ to the mobile device and executing it.

The DataWedge installation package is available from the Motorola Product Support site at http://support.symbol.com/support/product/DEV_SW_TOOLS.html.

Installing DataWedge on a PC

Run the DataWedge installation package on the PC. Follow the instructions provided by the installation wizard to complete the installation. The following files/folders are installed on the PC.

- Cab\DataWedge.CAB - DataWedge Cabinet file
- INI\DataWedge.ini - DataWedge configuration settings file
- DataWedgeInstaller.exe - Executable program to install DataWedge on a mobile device
- DataWedge Configuration Guide
- Remote Config\ - DataWedge Remote Configuration folder
- Readme.htm - Quick reference file
- WebUpdates.htm - DataWedge updates web location

Installing DataWedge on a Mobile Device

1. Establish a Microsoft ActiveSync® connection between host PC and mobile device.
2. On the host PC, go to Start > Programs > Motorola DataWedge > Install DataWedge to initiate the automatic installation process.
3. A screen displaying installation details appears on the mobile device.

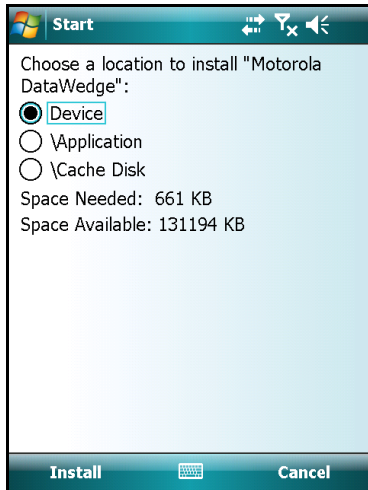


Figure 2-1 *Installation Location Details*

Select the preferred install location using the radio button and press **Install** to proceed with the installation of DataWedge on the mobile device.

4. Wait a few moments while DataWedge is installed to the mobile device. After a successful installation, a message window appears to announce that DataWedge is installed.

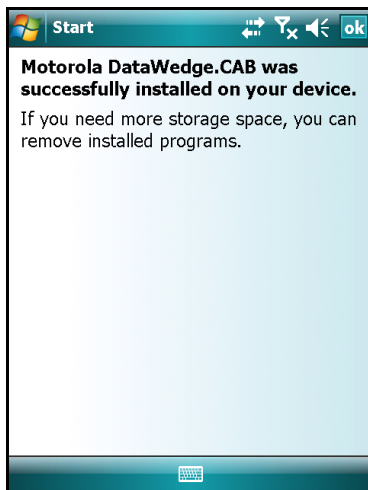


Figure 2-2 *Successful Installation Notification*

Tap ok to close the message window.

5. After the installation is completed DataWedge starts automatically.

Please refer to the *DataWedge Advanced Configuration Guide* for details on different options available for installing DataWedge on a Mobile Device.

DataWedge Tray Icon

DataWedge is launched on the mobile device upon successful installation. The DataWedge icon appears on the windows taskbar to indicate that DataWedge is in operation. Tap on the icon to open the DataWedge tray icon menu.

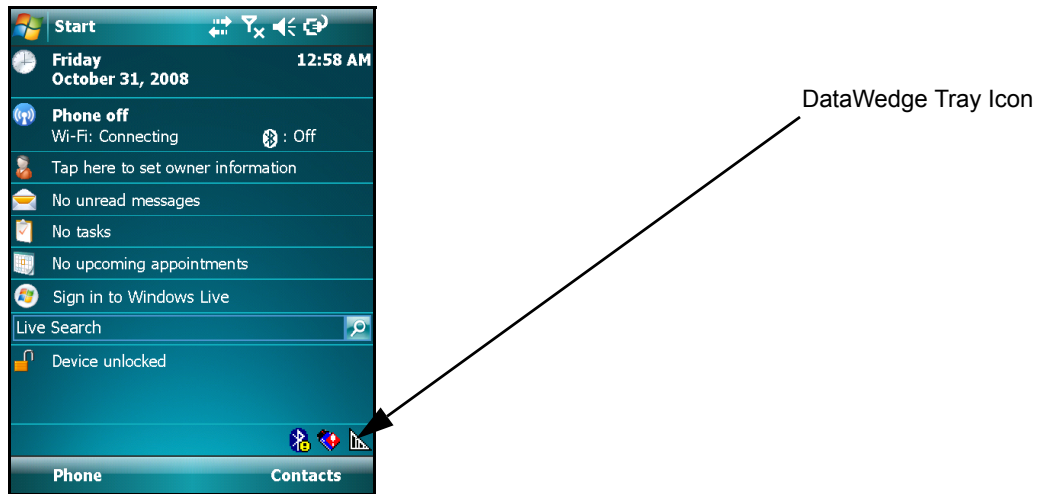


Figure 2-3 *Mobile Device Desktop (DataWedge Icon)*

DataWedge Tray Icon Menu

Use the tray icon menu to start/stop DataWedge, to access basic/advanced configuration modes and to terminate DataWedge activities on the mobile device.

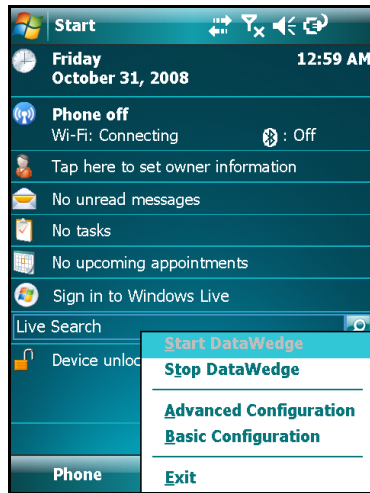


Figure 2-4 DataWedge Tray Icon Menu

- Select *Start DataWedge* to launch DataWedge on the mobile device.
- Select *Stop DataWedge* to stop DataWedge on the mobile device. When this option is selected, DataWedge can be launched again using the tray icon menu.
- Select *Advanced Configuration* to launch advanced configuration mode.
- Select *Basic Configuration* to launch basic configuration mode.
- Select *Exit* to close DataWedge on the mobile device. When this option is selected, DataWedge is shut down and the tray icon is hidden as well. To start DataWedge again use the Start Menu.

Uninstalling DataWedge

DataWedge can be uninstalled from the mobile device via the host PC or by using the Add/Remove Programs applet on the mobile device.

Remove DataWedge via Host PC

Method 1

1. Establish a Microsoft ActiveSync® connection between host PC and the mobile device.
2. On the host PC, go to Start > Programs > Motorola DataWedge > DataWedge Installer.
3. When the *Applications Already Installed* prompt appears, select **No** to move to the *Add/Remove Programs* window.

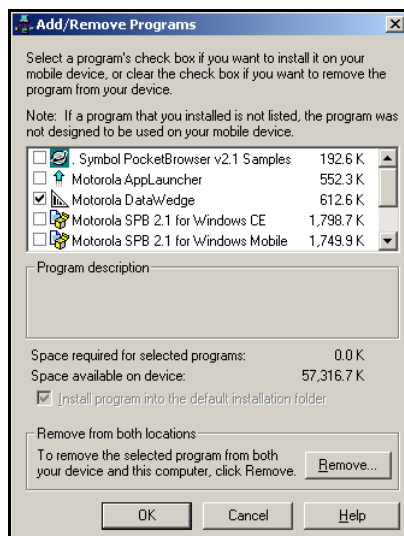


Figure 2-5 Add/Remove Programs Window

4. De-select the checkbox alongside Motorola DataWedge and press **OK** to remove DataWedge from the mobile device.

Method 2

1. Establish a Microsoft ActiveSync® connection between the mobile device and the host PC.
2. In the *Notification Area* of the host PC, right click the ActiveSync icon and select *Open Microsoft ActiveSync* option or alternatively, double-click the ActiveSync icon to open the Microsoft ActiveSync window.
3. In the Microsoft ActiveSync window go to Tools > Add/Remove Programs.
4. De-select the checkbox alongside Motorola DataWedge and press **OK** to remove DataWedge from the mobile device.

✓ **NOTE** Apart from the described methods, DataWedge can be uninstalled from both the host PC and from the mobile device by highlighting Motorola DataWedge from the list in the *Add/Remove Programs* window and pressing the **Remove** button under *Remove from both locations* panel. Press **OK** when the *Remove Application* dialog box appears, to confirm removal of DataWedge from the mobile device and the host PC. This option only removes the temporarily stored CAB file from the host PC and not the DataWedge program group which includes Readme, DataWedge Configuration Guide etc.

Chapter 3 Configuring Basic View

Introduction

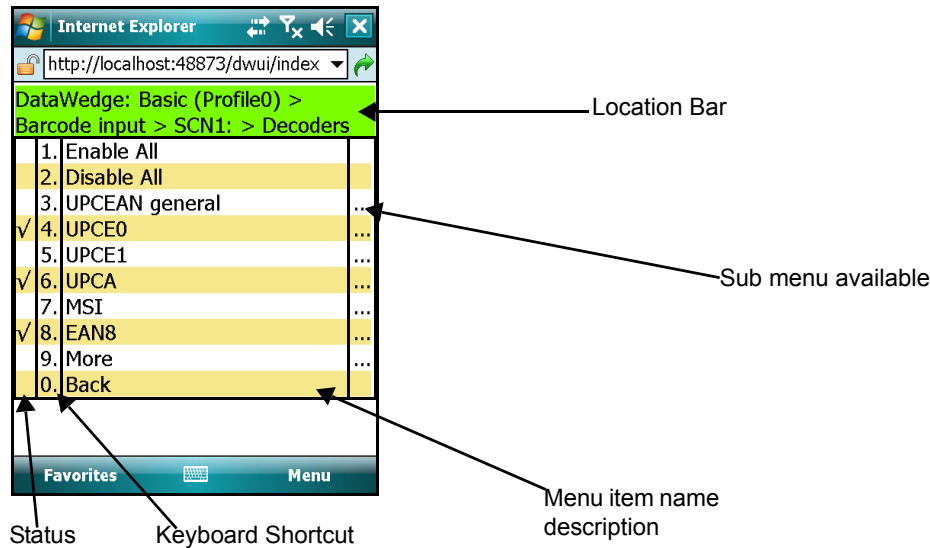
DataWedge configuration is handled through a browser based interface. It consists of a hierarchy of menus which can be navigated using the keypad or the touch-sensitive screen (if present). The DataWedge configuration settings are saved in XML files.

- ✓ **NOTE** DataWedge can be remotely configured via the PC using the Remote Configuration option available in DataWedge programs group. The configuration screens in DataWedge Remote Configuration procedure is identical to DataWedge configuration running on the mobile device, i.e. it is a PC emulation of DataWedge configuration. Refer to the *DataWedge Advanced Configuration Guide* for more details.

Navigate through the basic view menu items to access basic configuration options.

General Format of User Interface

The DataWedge configuration user interface (UI) has a number of elements. Running across the top of the page is a location bar, which indicates the current location within the menu hierarchy.



The menu item list is formatted into four columns. The first is a status column indicating whether the item is enabled or not, where applicable. The second column gives the keyboard shortcut for that menu item, enabling navigation of the menu without the need of touch screen input. Column three is the name/description of the menu item. The fourth column is a sub menu indicator that generally displays ellipses ("...") if a sub menu is available for that menu item. Access the sub menu by selecting that menu item.

The "0" item is universally used as the shortcut to navigate to the previous page. In the main menu only, the "0" item is used to exit from the configuration utility.

Launching Basic Configuration

To launch DataWedge basic configuration view, select *Basic Configuration* from the DataWedge tray icon menu.

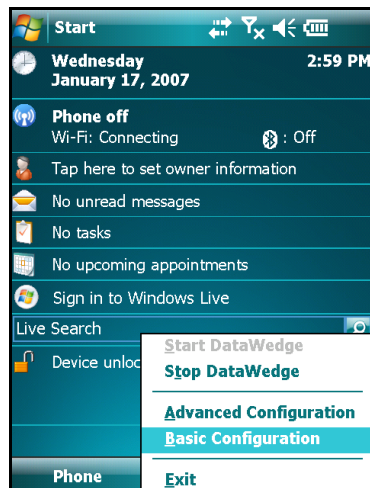


Figure 3-6 Launching Basic View (DataWedge Tray Icon Menu)

Main Configuration Menu

A main configuration menu is displayed on the screen when the Basic configuration is launched on the mobile device. This page consists of four menu items, namely *Barcode input*, *Keystroke output*, *Basic format* and *Exit*.

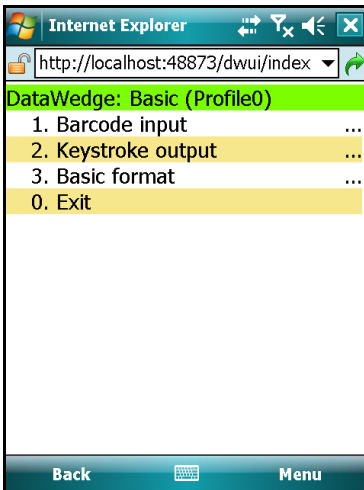


Figure 3-7 Main Configuration Menu Page

Use the appropriate keyboard shortcut or the touch screen to navigate through the main menu.

- Select *Barcode Input* to access the barcode plug-in configuration menu.
- Select *Keystroke output* to access the keystroke plug-in configuration menu.
- Select *Basic format* to access the basic format plug-in configuration menu.
- Select *Exit* to exit from DataWedge basic configuration mode.

✓ **NOTE** Do not press any buttons until configuration menu is fully loaded as it might hamper the loading process.

Configuring Barcode Input Plug-in

The barcode plug-in reads the captured data from barcode scanners and queues the data for processing.

Select *Barcode input* from the DataWedge basic configuration main menu to move to the *Barcode input* menu page.

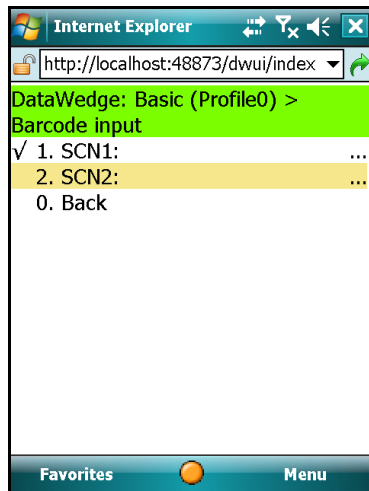


Figure 3-8 Scanner Device Selection Page

Use this page to select the scanner device. Because this page is the entry point to the barcode input plug-in configuration, moving back to the previous page, displays a dialog box prompting confirmation to save any changes made to the barcode input plug-in. Tap **OK** to save changes. Tap **Cancel** to revoke any changes.

When the scanner device is selected, DataWedge configuration moves to the input plug-in configuration main page where all configurable options for that scanner are listed.

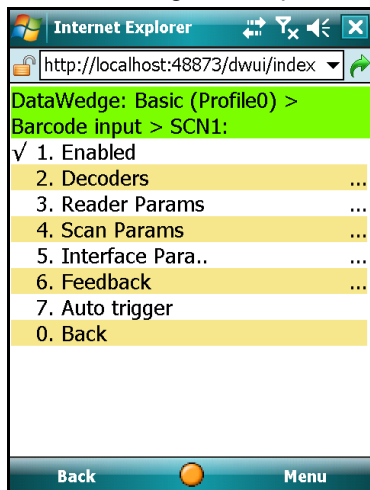


Figure 3-9 Barcode Plug-in Configuration Main Menu

The following is a brief description of the menu items on the scanner plug-in configuration page.

- Select *Enabled* to enable/disable the scanner. Disabling the scanner makes DataWedge basic configuration mode inoperable.

- Select *Decoders* to access the supported decoders for the scanner.
- Select *Reader Params* to access the reader parameters for the scanner. (See [Reader Parameters](#))
- Select *Scan Params* to access the scanner parameters for the scanner. (See [Scanner Parameters](#))
- Select *Interface Params* to access the interface parameters for the scanners. (See [Interface Parameters](#))
- Select *Feedback* to configure the notification options for the scanner. (See [Input Plug-in Feedback Settings](#))
- Select *Auto trigger* to enable/disable auto trigger mode for the scanner. When enabled, the scanner continuously reads barcodes. Use this feature for Motorola Micro Kiosks, such as the MK500. (See [Enable/Disable Auto Trigger Mode](#))



NOTE Use of this feature on a battery powered mobile device is not recommended because it can cause the battery to discharge more rapidly.

Enabling/Disabling the Scanner

Select *Enable* to enable the scanner. When the scanner is enabled, a tick (✓) is displayed alongside the *Enabled* item. To disable, select *Enable* again to toggle the Enabled state.

Configuring Scanner Decoders

Before using the scanner to capture data ensure that the required symbologies are enabled. Select the *Decoders* option from the scanner configuration menu to move to the page listing all decoders supported by the scanner.

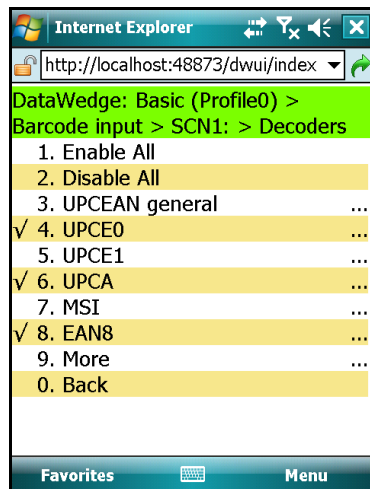


Figure 3-10 *Decoders List*

This menu page lists all the decoders supported by the scanner. Use the *More* option to navigate through the list to configure additional decoders.

- Select *Enable All* to enable all decoders for the selected barcode scanner.
- Select *Disable All* to disable all the decoders.

Configuring Decoders

Select the decoder name from the list, to navigate to the page containing the configurable parameters for that particular decoder.

Example

Configuring EAN8 Decoder

Select *EAN8* from the list to move to the EAN8 decoder configuration page.

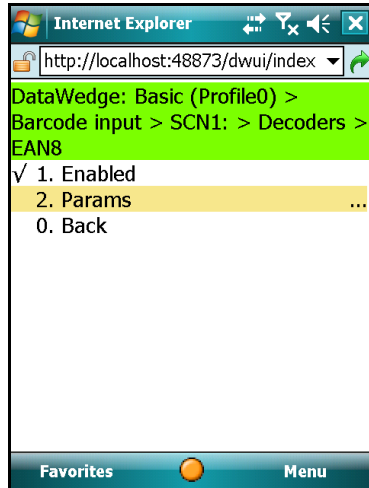


Figure 3-11 EAN8 Decoder Page

The *Enabled* option changes the enable/disable status of the EAN8 decoder. When enabled the scanner allows reading of EAN8 barcodes.

Select *Params* to configure additional parameters of the EAN8 decoder.

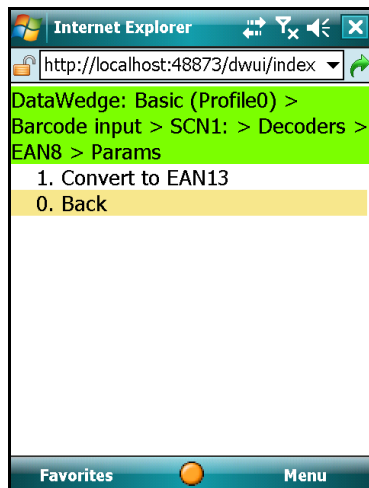


Figure 3-12 EAN8 Params Page

Use the *Convert to EAN13* option to enable/disable conversion of EAN8 barcodes to EAN13 barcodes. A tick (✓) is displayed when this option is enabled.

Configuring Reader Parameters

Select *Reader Params* from the scanner configuration menu to configure reader specific parameters.

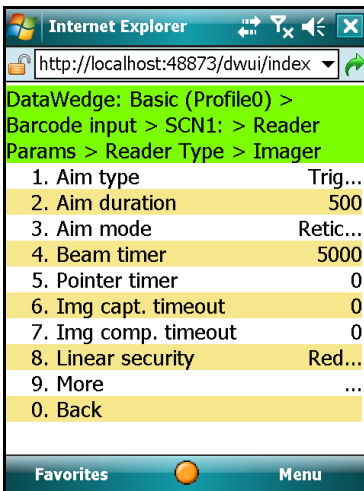


Figure 3-13 *Reader Parameters Page*

Reader Parameters

The *Reader Parameters* settings differ depending on the barcode reader type. See [Laser Scanner Reader Parameters](#) and [Imager Reader Parameters](#) for details.

Laser Scanner Reader Parameters**Table 3-1** Laser Scanner Reader Parameters

Reader Parameters	Laser Values	Description
Reader Type	Laser	Laser type scan engine is used.
Aim type	Trigger mode:	On/off controlled by the trigger.
	Trigger Hold mode:	Trigger can be released but it remains active for the specified period of time.
	Timed Release mode:	Activation stops after a specified period of time, even if the trigger is held.
Aim duration	0 - 60,000 ms	Sets the amount of time (0 - 60,000 ms in increments of 100 ms).
Aim mode	Dot, Slab, Reticle, None	Describes the aiming modes to use Dot – Projects a dot used for aiming Slab – Projects a line used for aiming Reticle – Projects an aiming pattern used for framing a barcode. None – set to none to disable this
Narrow beam	Enable, Disable	Sets the scan beam width to normal or narrow. Enable – Enable narrow beam Disable – Disable narrow beam (enable normal beam)
Raster mode	Smart	Creates a single scan line which opens vertically for PDF417 symbols using the Smart Raster feature. This feature auto detects the type of bar code presented and adjusts its pattern accordingly. This provides optimal performance on 1D, PDF417, and EAN/UCC.
	Cyclone	A scan pattern which decodes 1D symbologies in any orientation.
	None	Raster mode disabled.
	Open Always	Opens the laser to a full sized raster pattern. Decodes 1D and PDF417. NOTE Raster Mode is not supported on all devices.
Beam timer	0 - 60,000 ms	Sets the maximum amount of time that the laser remains on (0 - 60,000 ms in increments of 100 ms). A value of 0 sets the laser to stay on.
Control scan LED	Enable, Disable	Not supported, do not modify default setting.
Scan LED logic level	Enable, Disable	Not supported, do not modify default setting.
Klasse Eins enable	Enable, Disable	Not supported, do not modify default setting.
Bidir. redundancy	Enable, Disable	Sets the read direction for the bar code redundancy. Bidirectional reads in both directions.

Table 3-1 Laser Scanner Reader Parameters

Reader Parameters	Laser Values	Description
Linear security level Linear Sec (Laser only)		Sets the number of times a bar code is re-read to confirm an accurate decode.
	All twice:	All twice: Two times read redundancy for all bar codes.
	All thrice:	All thrice: Three times read redundancy for all bar codes.
	Long and Short:	Long and Short: Two times read redundancy for long bar codes, three times for short bar codes.
	Redundancy + length:	Redundancy + length: Two times read redundancy based on redundancy flags and code length.
	Short or Codabar	Short or Codabar: Two times read redundancy if short bar code or CODABAR.
Pointer timer	0 - 60,000 ms	Sets the maximum amount of time that the pointer remains on (0 - 60,000 ms in increments of 100 ms). A value of 0 sets the pointer to stay on.
Raster height	0-100 in.	Sets the Raster Height from 0 to 100 inches in increments of 5 in. Raster Height is not supported on all devices.
DBP Mode		Describes what type of Digital Bar Pulse (DBP) is being produced by the scan engine.
	Normal	Normal – tells the engine to produce normal DBP.
	Composite	Composite – tells the engine to produce composite DBP, which is 2 different sets of DBP data multiplexed together for better decode performance. Note: If the device does not support I2C or if using an older engine the default value for DBP Mode is Normal. An attempt to change this mode to Composite results in an E_SCN_NOTSUPPORTED error.

Imager Reader Parameters**Table 3-2** Imager Reader Parameters

Reader Parameters	Imager Values	Description
Reader Type	Imager	Imager type engine is used.
Aim type	Trigger mode:	On/off controlled by the trigger.
	Timed hold mode:	Trigger can be released but it remains active for the specified period of time.
	Timed Release mode:	Activation stops after a specified period of time, even if the trigger is held.
	Presentation	Special mode enables scanning when motion is detected in front of the imager. (Currently only supported by MK500)
Aim duration	0 - 60,000 ms	Sets the amount of time (0 - 60,000 ms in increments of 100 ms).
Aim mode	Dot, Slab, Reticle, None	Describes the aiming modes to use Dot – Projects a dot used for aiming Slab – Projects a line used for aiming Reticle – Projects an aiming pattern used for framing a barcode. None – set to none to disable this NOTE Both Dot and Slab options are invalid for imager, thus if selected the setting is overridden to reticle mode.
Beam timer	0 - 60,000 ms	Sets the maximum amount of time that the laser remains on (0 - 60,000 ms in increments of 100 ms). A value of 0 sets the laser to stay on.
Pointer timer	0 - 60,000 ms	Sets the maximum amount of time that the Pointer Timer remains on (0 - 60,000 ms in increments of 100 ms). A value of 0 sets the Pointer Timer to stay on. NOTE This parameter is not supported on all devices.
Img capt. timeout	0 - 60,000 ms	Sets the maximum amount of time for the Image Capture Timeout (0 - 60,000 ms in increments of 100 ms). A value of 0 sets the Image Capture Timeout to stay on. NOTE This parameter is not supported on all devices.
Img comp. timeout	0 - 60,000 ms	Sets the maximum amount of time for the Image Compress Timeout (0 - 60,000 ms in increments of 100 ms). NOTE Image Compress Timeout is not supported on all devices.

Table 3-2 *Imager Reader Parameters*

Reader Parameters	Imager Values	Description
Linear security		Sets the number of times a bar code is read to confirm an accurate decode.
	All twice:	All twice: Two times read redundancy for all bar codes.
	All thrice:	All thrice: Three times read redundancy for all bar codes.
	Long and Short:	Long and Short: Two times read redundancy for long bar codes, three times for short bar codes.
	Short or Codabar	Short or Codabar: Two times read redundancy if short bar code or CODABAR.
Focus mode	Fixed, Auto	Fixed mode is the only supported focus mode.
Focus position	Far, Near	Specifies the Fixed setting, focus position for Far is 9 inches and focus position for Near is 5 inches.
Poor quality mode	Enable, Disable	This parameter allows poor quality 1D bar codes to be read, BUT adversely affecting the overall decoding performance. Enable – Enables poor quality decoding for 1D barcodes. Disable – Disables poor quality decoding for 1D barcodes.
Picklist mode	Disabled, Enabled/HW reticule, Software reticule	This parameter allows the imager to decode only the bar code that is directly under the cross-hair/reticule (+) part of the AIM pattern. This feature is most useful in applications where multiple bar codes may appear in the field of view during a decode session and only one of them is targeted for decode. When enabled, bPicklistMode overrides dwAimMode if no aiming is chosen and use the AIM_MODE_RETICLE mode. When enabled, bPicklistMode may adversely affect overall decoding performance. Disabled – Disables picklist mode, so any bar code within the field of view can be decoded. Enable/HW reticule – Enables picklist mode, so only the bar code under the cross-hair can be decoded. Software reticule - Enables picklist mode, so only the bar code under the cross-hair can be decoded. In this mode the reticule is seen on the viewfinder as oppose to on the barcode surface. Especially used with Camera Scan.

Table 3-2 *Imager Reader Parameters*

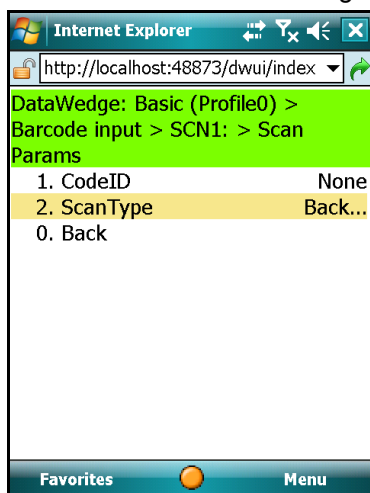
Reader Parameters	Imager Values	Description
DPM Mode	Enable, Disable	This parameter allows Direct Part Marking (DPM) bar codes to be read but may adversely affect overall decoding performance. DPM is a way of stamping bar codes directly on physical objects. Support for this feature is available on DPM enabled mobile computers only. If this feature is not available and user attempts to enable it, an error (E_SCN_NOTSUPPORTED) results. Enable – Enables decoding of DPM bar codes. Disabled – Disables decoding of DPM bar codes. NOTE This feature cannot be turned on in conjunction with Picklist as both these modes are mutually exclusive. An attempt to turn on both results in an error (E_SCN_NOTSUPPORTED).
Illumination mode	Auto, Always off, Always On	Illumination modes to use. Possible values are: Auto Illumination – In this mode the auto-exposure algorithms decides whether illumination is required or not. Always on – In this mode external illumination is always on. Always off – In this mode external illumination is always off.
VF left pos.	0 – 600	This setting displays the top left X coordinate of the viewfinder window.
VF top pos.	0 – 800	This setting displays the top left Y coordinate of the viewfinder window.
VF right pos.	0 – 600	This setting displays the bottom right X coordinate of the viewfinder window.
VF bottom pos.	0 – 800	This setting displays the bottom right Y coordinate of the viewfinder window.
VF mode	Disabled, Enabled, Static reticule, Dynamic reticule	This setting displays the Viewfinder modes supported for scanning. Possible values are: Disable - Viewfinder is not displayed during aiming and scanning. Enabled - Only Viewfinder is enabled. Static Reticule - Displays the Viewfinder as well as draws a red reticule in the center of the screen which helps tracking the barcode. Dynamic Reticule - Displays the Viewfinder as well as draws a red reticule in the center of the image. If the barcode in the image is 'decodable' the reticule turns green to indicate this.

Table 3-2 Imager Reader Parameters

Reader Parameters	Imager Values	Description
VF feedback	Disabled, Enabled, Reticule	This parameter allows selection of the different feedback parameters on a successful decode. Possible values are: Disabled - This mode disables any visual feedback on a successful decode. Enabled - This mode displays the last image that successfully decoded. The duration for which the image is displayed can be set by the Viewfinder feedback time. Reticule - This mode displays the last image that successfully decoded and also draws a reticule in the center of the image.
VF feedback time	0 - 60,000 ms	This displays the Time for which the visual display selected by Viewfinder feedback mode. For more information Please refer <i>Motorola Enterprise Mobility Developer Kit for C Help</i> .
Inverse 1d Mode	Disabled, Enabled, Auto	This parameter allows the user to select decoding on inverse 1D barcodes. Disabled - Disables decoding of inverse 1D symbologies. Enabled - Enables decoding of only inverse 1D symbologies. Auto - Allows decoding of both positive as well as inverse 1D symbologies.

Configuring Scan Parameters

Select *Scan Params* to configure the scan parameters.

**Figure 3-14** Scanner Parameter Configuration Page

Use this menu to access and configure the scan parameters.

- Select *CodeID* to specify the CodeID.

- Select *Scan Type* to specify the type of Code ID to be reported.

See [Scan Parameters](#) for more details.

Scanner Parameters

Below table lists all the scan parameters.

Table 3-3 *Scan Parameters*

Scan Parameters	Values	Description
Code ID Type	None	Default setting. No prefix
	Symbol	A Symbol defined single character prefix.
	AIM	A standard based three character prefix.
Scan Type	Foreground	Foreground reads combine only with other foreground reads and preempt background reads.
	Background	The scan takes place in the background, but only if no foreground reads are pending.
	Monitor	No scanning is requested, but if scanning is initiated by another application, a monitor read receives a copy (if the code type is appropriate).



NOTE By default, the Scan Type is set to 'Background'. This allows DataWedge to share the scanner with other scan enabled applications. Setting this parameter to 'Foreground' is not recommended as this may interfere with other scan enabled applications or vice versa.

Configuring Interface Parameters

Select *Interface Params* from the scanner configuration menu to set the interface parameters.

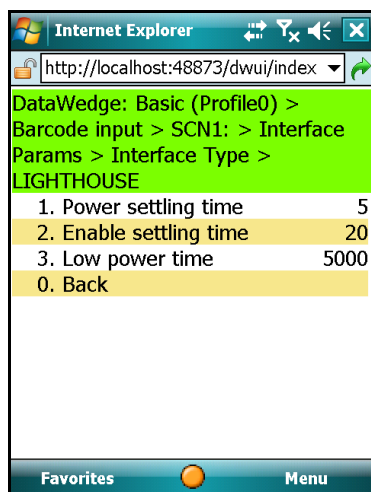


Figure 3-15 *Interface Parameter Page*

Select a parameter from the list to modify the default value assigned to it.

See [Interface Parameters](#) for more details.

Interface Parameters

Table 3-4 *Interface Parameters*

Interface Parameter	Values
Interface Type	Lighthouse, Camscan
Power Settle Time	0 - 100 ms
Enable Settle Time	0 - 100 ms
Low Power Time	0 - 60,000 ms

Input Plug-in Feedback Settings

Select the *Feedback* option from the scanner configuration menu to navigate to the feedback settings page where the feedback parameters can be configured.

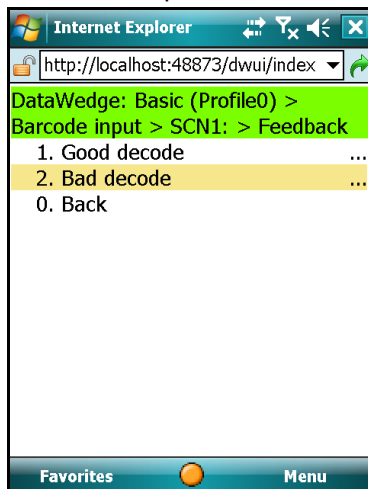


Figure 3-16 *Feedback Selection Page*

Use DataWedge configuration pages to configure the feedback settings for the selected input plug-in.

The scanner plug-in has two events which feedback settings need to be configured. The feedback mechanism uses the LED, Beeper and Wave Files to provide feedback to the user.

- Select *Good Decode* to configure feedback settings for a successful decode.
- Select *Bad Decode* to configure feedback settings for a unsuccessful decode.

Select either Good or Bad decode from the list to move to the corresponding page where the feedback options are listed.

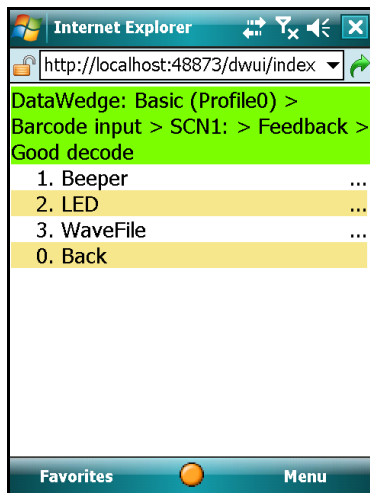


Figure 3-17 Feedback Module Selection Page

Select a feedback option from the list to configure it.

- Select the *Beeper* option to access and configure Beeper feedback.
- Select the *LED* option to access and configure LED feedback.
- Select the *WaveFile* option to access and configure Wave File feedback.

Configuring Beeper Feedback Settings

Select *Beeper* to configure beeper feedback settings.

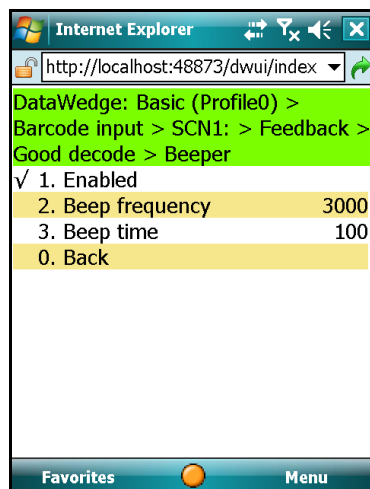


Figure 3-18 Beeper Configuration Page

- Select *Enabled* option to enable or disable the beeper feedback. When enabled, a tick (✓) is displayed alongside the *Enabled* item. To disable, select Enable again to toggle the Enabled state.
- Select *Beep frequency* option to set the Beep frequency. A form appears allowing the beep frequency to be changed. Enter the desired value and press **Save** to save.

- Select *Beep time* option to set the beep duration. A form appears allowing the beep duration to be changed. Enter the desired time (in milliseconds) and press **Save** to save.

Configuring LED Feedback Settings

Select LED to configure the LED feedback settings.

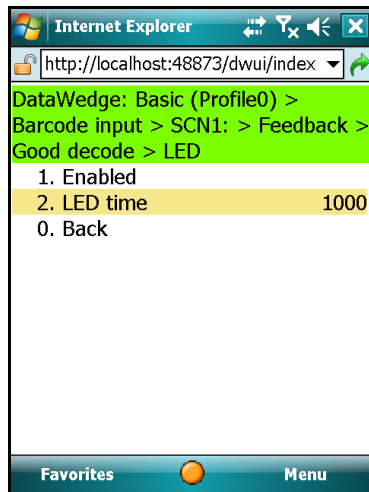


Figure 3-19 LED Configuration Page

- Select *Enabled* option to enable or disable the LED feedback.
- Select *LED time* option to set the LED time. Enter the time duration (in milliseconds) in the form that appears and press **Save** to save.

Configuring WAV Feedback Settings

Select WaveFile to configure WAV feedback setting.

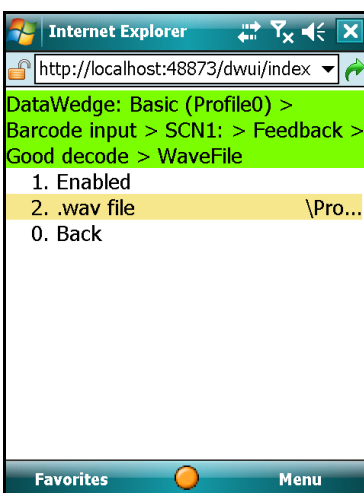


Figure 3-20 Wave File Configuration Page

Select *Enabled* option to enable or disable the WaveFile feedback.

Select *.wav file* option to specify the WAV file to be used. Enter the path/filename of the .wav file in the form that appears and press **Save** to save.

Enable/Disable Auto Trigger Mode

From the barcode plug-in configuration main menu, select *Auto trigger* to enable auto trigger mode for the scanner. When this feature is enabled, the scanner is activated when movement is detected beneath it and automatically scans barcodes. Use this feature for Motorola Micro Kiosks, such as the MK2200.



NOTE Use of this feature on a battery powered mobile device is not recommended because it can cause the battery to discharge more rapidly

By default this feature is disabled on DataWedge. When enabled a tick (✓) is displayed alongside Auto trigger menu item. To disable, select the menu item again to toggle the state.



NOTE Use of this feature with the camera is not recommended because the Viewfinder is given precedence over the other foreground applications and therefore, DataWedge configuration interface can be obscured. Use the DataWedge Remote Configuration to change DataWedge settings whilst in this mode.

Configuring Keystroke Output Plug-in

The Keystroke output plug-in dispatches data to the foreground application in the form of keystrokes. The Keystroke plug-in can receive data either directly from input plug-ins or via process plug-ins.

Select *Keystroke output* to move to Keystroke configuration main menu page.

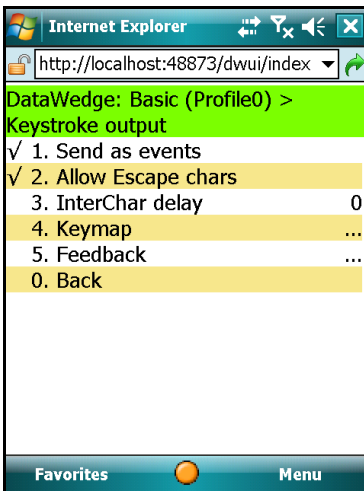


Figure 3-21 Keystroke Plug-in Configuration Page

- Select *Send as events* to enable sending keystrokes as keyboard events. When enabled, a tick (✓) is displayed alongside Send as events menu item. To disable this feature, select the menu item again to toggle the state. When disabled, Keystrokes are sent to the foreground application as messages.
- Select *Allow Escape chars* to enable DataWedge to recognise escape sequences in the incoming data and convert them to keystrokes. A tick (✓) is displayed alongside the menu item when this feature is enabled. To disable this feature, select the menu item again. When disabled, escape sequences are sent as data.
- Select *InterChar delay* option to specify the delay to be inserted between each keystroke that is sent.
- Select *Keymap* to specify the keymap settings.
- Select *Feedback* to specify the feedback settings for events handled by the output plug-in.

This page is the entry point to the keystroke output plug-in configuration. When moving back to the previous page a dialog box is displayed, prompting confirmation to save any changes made to the keystroke output plug-in. Press **OK** to save changes. Press **Cancel** to revoke any changes.

Allow Escape Characters

The *Allow Escape chars* option configures the Keystroke plug-in to recognize escape sequences in the incoming data buffer. When this option is enabled DataWedge can process the escape characters in an incoming data stream and also send escape characters to the foreground application via the Keystroke plug-in. Disabling this option causes DataWedge to leave escape characters unchanged. For example if a barcode contains characters "\r" and if Allow Escape chars is enabled, then DataWedge replaces the "\r" with a carriage return character.

However, if Allow Escape chars property is disabled DataWedge treats the "\r" as regular characters, leaving them unchanged. The above scenario is also true for data modifications done via Basic format process plug-in.

Table 3-5 *Escape Sequences Supported by DataWedge*

Escape Sequence	Description
\b	Backspace
\f	Form feed
\n	New line
\r	Carriage return
\S	Sticky key
\t	Horizontal tab
\u hhhh	Unicode character in hexadecimal notation.
\v hh	Virtual key represented in hexadecimal notation
\x hh	ASCII character in hexadecimal notation

Inter Character Delay

The inter character delay is the delay (in milliseconds) that is inserted between the keystrokes that are sent.

Select the *InterChar delay* option to move to a dialog where the inter character delay can be specified.

Use the mobile device keypad or the onscreen keyboard to enter the inter character delay and press **Save** to save.

Configuring Keymap Settings

The keymap configuration is used to translate characters from the incoming data to alternative characters before sending to the foreground application.

Select *Keymap* to move to keymap configurations page.

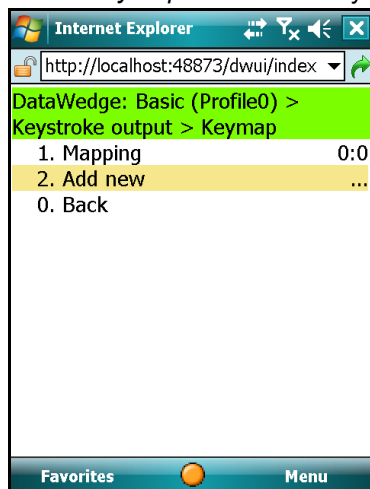


Figure 3-22 *Keymap Configuration Page*

Adding a Keymap

Select *Add new* to add a new key-mapping. A new option titled *Mapping* is added to the keymap configuration page.

Select the *Mapping* option to configure the new keymap.

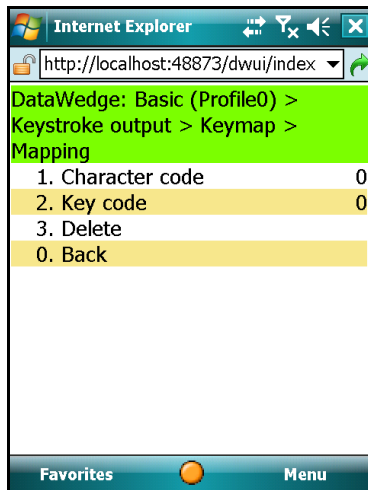


Figure 3-23 Key Mapping Main Page

- Select *Character code* to specify the ASCII value of the incoming character.
- Select *Key code* to specify the virtual key value of the alternate character.
- Select *Delete* to delete the selected key mapping.

Modifier Keys

Modifier keys are special keys that modify the normal action of another key, when two are pressed in combination. For example, <Alt> + <F4> in Microsoft Windows is used to close the program in a active window. By themselves, modifier keys usually does nothing. The most widely used modifier keys are Ctrl, Shift and Alt keys.

Table 3-6 Modifier Key Values

Modifier Key	Key Value (Decimal)	Key Value (Hexadecimal)
Shift	256	100
Ctrl	512	200
Alt	1024	400

Key Mapping Examples

The examples below explains the procedure of setting keymaps.

Example1

The following example describes how to configure the Keystroke plug-in to simulate SHIFT+8 for left round bracket "(" instead of SHIFT+9 which is the default (101 keyboard) mapping. This is a typical scenario encountered when using an application via Remote Desktop which uses a 106 keyboard (Japanese Keyboard) and scans a barcode containing an open round bracket.

1. Use the *Character code* option to specify the incoming character code that needs to be altered by the Keystroke plug-in.

In the text box enter the character code for open round bracket as decimal 40 (40 is the decimal representation of "(" according to the ASCII table) and press the **Save** button.

2. Use the *Key code* option to enter the outgoing key code combination and press **Save**.

Since the outgoing key code is a combination of two keys (SHIFT+8) it is represented as a the sum of "SHIFT" and "8"

The virtual key value in decimal format for Shift key is 256 (See [Modifier Key Values](#)). The virtual key value of character "8" is 56 (See [Virtual Key Codes](#)). Therefore the Key code is;

$$\text{Shift} + 8 = 256 + 56 = 321$$

Refer to [Virtual Key Codes](#) for key value information. Refer to the [ASCII Table](#) for character code values.

Output Plug-in Feedback Settings

Select *Feedback* from the Keystroke plug-in configuration page to move to *Feedback* menu page. Use the output plug-in feedback configuration page to set feedback properties for the Keystroke plug-in.

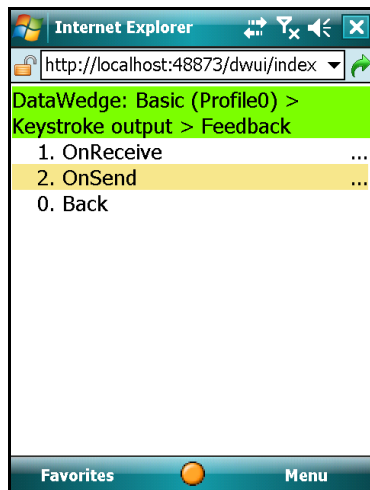


Figure 3-24 Feedback Main Menu Page

- Select *OnReceive* to configure feedback settings for data receive events which are triggered when the Keystroke plug-in receives data.
- Select *OnSend* to configure the feedback settings when the Keystroke plug-in sends data.

Configuring Output Plug-in Feedback Settings

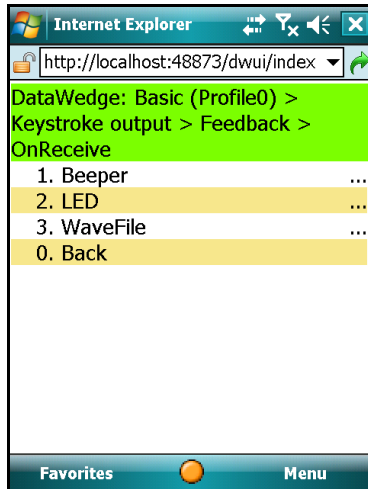


Figure 3-25 Keystroke Feedback Options

The feedback configuration options for the output plug-in are the same as those for the input plug-in. Please refer [Input Plug-in Feedback Settings](#) for additional details.

Configuring Basic Format Process Plug-in

Select *Basic format* from the main menu to configure the basic format process plug-in. When selected, DataWedge configuration moves to the basic format plug-in main menu page.

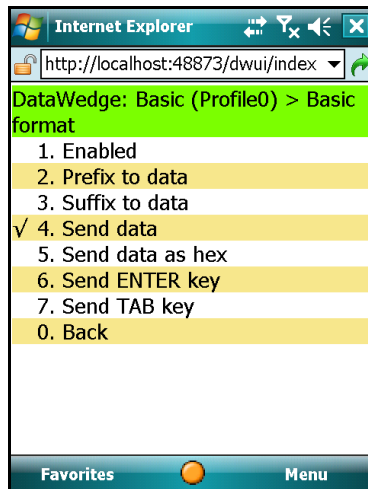


Figure 3-26 Basic Format Process Plug-in Configuration Menu

- Select *Enabled* option to enable or disable the Basic format process plug-in. When enabled, a tick (✓) is displayed alongside Enabled menu item. To disable, select the menu item again to toggle the state.
- Select *Prefix to data* to add a string to the beginning of the data.
- Select *Suffix to data* to add a string to the end of the data.

- Select *Send data* to transfer the captured data to the foreground application. Disabling this option prevents the actual data from been transmitted. The prefix and suffix strings, if present, are still transmitted even if this option is disabled.
- Select *Send data as hex* to send the data in hexadecimal format.
- Select *Send ENTER key* to append an enter character to the processed data.
- Select *Send TAB key* to append a tab character to the processed data.

Special Characters Supported by Basic format

The Basic format process plug-in supports the following special characters.

- Standard Escape Characters (See- [Escape Sequences Supported by DataWedge](#))
- Virtual Keys (See - [Virtual Key Codes](#))
- Hex representation of ASCII characters (See - [ASCII Table](#))
- Unicode Characters
- Sticky Keys (See [Sticky Key Definitions](#))

Sticky Keys

The format for sticky keys is defined as follows;

`\S(C|A|S)x[0..*]\E(C|A|S)`

The definitions of the sticky keys are described below..

Table 3-7 *Sticky Key Definitions*

Key Code	Description
\S, \s	Start sticky key
(C c)	C = CTRL
(A a)	A = ALT
(S s)	S = SHIFT
x[0..*]	0 or more character keys
\E, \e	End Sticky key.

When using sticky keys, use lower case characters to define key combinations. The key combination meanings may change the result depending on the characters used. For example, to depict CTRL+a, use \Sca or \SCa. If \SCA is used DataWedge emulates the key combination as CTRL+SHIFT+A which does not yield the required result.



NOTE In order to allow the escape characters to be supported, enable the "Allow Escape chars" option in the Keystroke output plug-in.

Basic Format Examples

Example1 - Displaying output data Line-by-line

The example below describes how to configure the Basic format process plug-in to alter the output data to be displayed line-by-line as shown.

Start

1234567890

End

For the Prefix, the word "Start" is followed by \r which inserts a carriage return before the data.

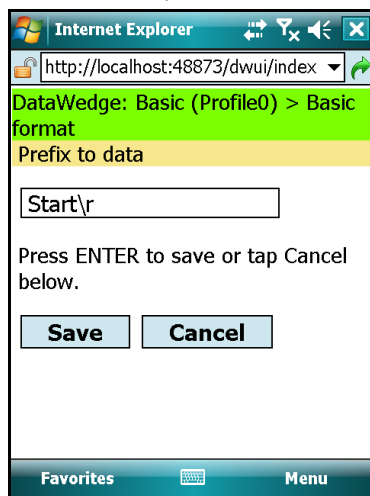


Figure 3-27 Prefix Setting

For the Suffix a \r is set before the word "End" which means the data is followed by a carriage return and then the word "End".

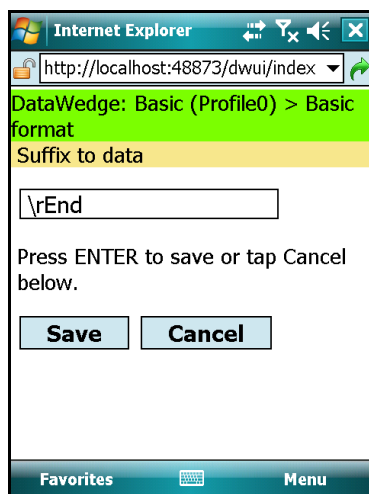


Figure 3-28 Suffix Setting

Example2 - Sending a linefeed after Data

A linefeed can be generated by typing CTRL+j on a keyboard. To emulate this in DataWedge, enter the Suffix string as \Scj\Ec.

Example3 - Fill a text field & press the OK button in a form

Where a form consists of a text input box and a submit button, DataWedge can be set to populate the text input box, then set focus to and press the button. This achieved by setting the Suffix as \Scim\Ec to simulate Tab and Enter.

Example4 - Make all characters uppercase

If the input data consists purely alpha characters (i.e. a-z), these can be converted to upper case simply by holding down the SHIFT keys while sending the characters. To achieve this effect in DataWedge, set the Prefix as \Ss and Suffix as \Es. This converts all lower case alpha characters in the data to uppercase.

Example5 - Open an MS Word document on a remote computer and print the data

Connect to a remote PC via Remote Desktop and launch MS Word. Set the prefix as \Safn\Es. This opens a new MS Word document and enter the data into the document.

To enter the data and automatically print that data afterwards, set the suffix as \Safp\Ea\Scm\Ec. After entering the data this invokes the MS Word File menu, selects Print and presses the enter key causing the document to be printed.

Appendix A Useful Information

Special Scenarios

Preventing Data Loss in Remote Desktop

When using the Remote Desktop application to run an application on a remote Windows server, certain conditions (such as a slow connection) can cause occasional keystrokes to be ignored by Remote Desktop or by the application it is running.

In this case, specify a value of 20ms or more for the InterChar delay.

Similarly, some applications may have trouble dealing with large amounts of rapidly occurring keystrokes, as might be generated when scanning barcodes such as PDF417. Setting a suitable InterChar delay can mitigate this problem.

Virtual Key Codes

The following table lists virtual key codes for a generic keyboard.

Table A-1 *Virtual Key Codes*

Key	Key Value (Decimal)	Key Value (Hexadecimal)
0	48	30
1	49	31
2	50	32
3	51	33
4	52	34
5	53	35
6	54	36
7	55	37

Table A-1 *Virtual Key Codes*

Key	Key Value (Decimal)	Key Value (Hexadecimal)
8	56	38
9	57	39
A	65	41
B	66	42
C	67	43
D	68	44
E	69	45
F	70	46
G	71	47
H	72	48
I	73	49
J	74	4A
K	75	4B
L	76	4C
M	77	4D
N	78	4E
O	79	4F
P	80	50
Q	81	51
R	82	52
S	83	53
T	84	54
U	85	55
V	86	56
W	87	57
X	88	58
Y	89	59
Z	90	5A
Space	32	20
Escape	27	1B

Table A-1 *Virtual Key Codes*

Key	Key Value (Decimal)	Key Value (Hexadecimal)
;	186	BA
=	187	BB
,	188	BC
-	189	BD
.	190	BE
/	191	BF
`	192	C0
[	219	DB
\	220	DC
]	221	DD
'	222	DE
F1	112	70
F2	113	71
F3	114	72
F4	115	73
F5	116	74
F6	117	75
F7	118	76
F8	119	77
F9	120	78
F10	121	79
F11	122	7A
F12	123	7B
Page Up	33	21
Page Down	34	22
End	35	23
Home	36	24
Left	37	25
Up	38	26
Right	39	27

Table A-1 *Virtual Key Codes*

Key	Key Value (Decimal)	Key Value (Hexadecimal)
Down	40	28
Insert	45	2D
Delete	46	2E
Backspace	8	08
Tab	9	09
Print Screen	44	2C
Shift	16	10
Ctrl	17	11
Caps Lock	20	14

ASCII Table

Regular ASCII Chart (character codes 0 - 127)

000d 00h	(nul)	016d 10h	► (dle)	032d 20h	sp	048d 30h	0	064d 40h	@	080d 50h	P	096d 60h	`	112d 70h	p
001d 01h	☉ (soh)	017d 11h	◄ (dcl)	033d 21h	!	049d 31h	1	065d 41h	A	081d 51h	Q	097d 61h	a	113d 71h	q
002d 02h	● (stx)	018d 12h	↑ (dc2)	034d 22h	"	050d 32h	2	066d 42h	B	082d 52h	R	098d 62h	b	114d 72h	r
003d 03h	♥ (etx)	019d 13h	!! (dc3)	035d 23h	#	051d 33h	3	067d 43h	C	083d 53h	S	099d 63h	c	115d 73h	s
004d 04h	♦ (eot)	020d 14h	‡ (dc4)	036d 24h	\$	052d 34h	4	068d 44h	D	084d 54h	T	100d 64h	d	116d 74h	t
005d 05h	♣ (enq)	021d 15h	§ (nak)	037d 25h	%	053d 35h	5	069d 45h	E	085d 55h	U	101d 65h	e	117d 75h	u
006d 06h	▲ (ack)	022d 16h	■ (syn)	038d 26h	&	054d 36h	6	070d 46h	F	086d 56h	V	102d 66h	f	118d 76h	v
007d 07h	● (bel)	023d 17h	‡ (etb)	039d 27h	'	055d 37h	7	071d 47h	G	087d 57h	W	103d 67h	g	119d 77h	w
008d 08h	▣ (bs)	024d 18h	↑ (can)	040d 28h	(	056d 38h	8	072d 48h	H	088d 58h	X	104d 68h	h	120d 78h	x
009d 09h	(tab)	025d 19h	↓ (em)	041d 29h	)	057d 39h	9	073d 49h	I	089d 59h	Y	105d 69h	i	121d 79h	y
010d 0Ah	(lf)	026d 1Ah	◻ (eof)	042d 2Ah	*	058d 3Ah	:	074d 4Ah	J	090d 5Ah	Z	106d 6Ah	j	122d 7Ah	z
011d 0Bh	␣ (vt)	027d 1Bh	← (esc)	043d 2Bh	+	059d 3Bh	;	075d 4Bh	K	091d 5Bh	[	107d 6Bh	k	123d 7Bh	{
012d 0Ch	♀ (np)	028d 1Ch	~ (fs)	044d 2Ch	,	060d 3Ch	<	076d 4Ch	L	092d 5Ch	\	108d 6Ch	l	124d 7Ch	
013d 0Dh	(cr)	029d 1Dh	↔ (gs)	045d 2Dh	-	061d 3Dh	=	077d 4Dh	M	093d 5Dh	]	109d 6Dh	m	125d 7Dh	}
014d 0Eh	↓ (so)	030d 1Eh	▲ (rs)	046d 2Eh	.	062d 3Eh	>	078d 4Eh	N	094d 5Eh	^	110d 6Eh	n	126d 7Eh	~
015d 0Fh	○ (si)	031d 1Fh	▼ (us)	047d 2Fh	/	063d 3Fh	?	079d 4Fh	O	095d 5Fh	_	111d 6Fh	o	127d 7Fh	◻

Extended ASCII Chart (character codes 128 – 255; Codepage 850)

128d 80h	Ç	144d 90h	É	160d A0h	á	176d B0h	⌚	192d C0h	Ł	208d D0h	Đ	224d E0h	Ó	240d F0h	-
129d 81h	ü	145d 91h	æ	161d A1h	í	177d B1h	⌚	193d C1h	ł	209d D1h	đ	225d E1h	ó	241d F1h	±
130d 82h	é	146d 92h	⌚	162d A2h	ó	178d B2h	⌚	194d C2h	Ł	210d D2h	Đ	226d E2h	ô	242d F2h	±
131d 83h	â	147d 93h	ô	163d A3h	ú	179d B3h		195d C3h	ł	211d D3h	đ	227d E3h	ò	243d F3h	¾
132d 84h	ä	148d 94h	ö	164d A4h	ñ	180d B4h	†	196d C4h	—	212d D4h	ē	228d E4h	ó	244d F4h	¶
133d 85h	à	149d 95h	ð	165d A5h	Ñ	181d B5h	†	197d C5h	†	213d D5h	†	229d E5h	ô	245d F5h	§
134d 86h	â	150d 96h	û	166d A6h	®	182d B6h	†	198d C6h	â	214d D6h	†	230d E6h	μ	246d F6h	+
135d 87h	ç	151d 97h	ù	167d A7h	º	183d B7h	†	199d C7h	Ä	215d D7h	†	231d E7h	þ	247d F7h	
136d 88h	ê	152d 98h	ÿ	168d A8h	¿	184d B8h	©	200d C8h	Ł	216d D8h	†	232d E8h	þ	248d F8h	
137d 89h	ë	153d 99h	ÿ	169d A9h	®	185d B9h	†	201d C9h	†	217d D9h	†	233d E9h	ÿ	249d F9h	~
138d 8Ah	è	154d 9Ah	ÿ	170d AAh	¬	186d BAh	†	202d CAh	†	218d DAh	†	234d EAh	ÿ	250d FAh	•
139d 8Bh	ï	155d 9Bh	ø	171d ABh	½	187d BBh	†	203d CBh	†	219d DBh	†	235d EBh	ÿ	251d FBh	1
140d 8Ch	í	156d 9Ch	£	172d Ach	¼	188d BCh	†	204d CCh	†	220d DCh	†	236d ECh	ÿ	252d FCh	2
141d 8Dh	ì	157d 9Dh	Ø	173d ADh	î	189d BDh	†	205d CDh	†	221d DDh	†	237d EDh	ÿ	253d FDh	3
142d 8Eh	Ä	158d 9Eh	×	174d AEh	„	190d BEh	†	206d CEh	†	222d DEh	†	238d EEh	†	254d FEh	■
143d 8Fh	Å	159d 9Fh	f	175d AFh	»	191d BFh	†	207d CFh	†	223d DFh	†	239d EFh	†	255d FFh	

Hexadecimal to Binary

0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

Groups of ASCII-Code in Binary

Bit 6	Bit 5	Group
0	0	Control Characters
0	1	Digits and Punctuation
1	0	Upper Case and Special
1	1	Lower Case and Special

Figure A-1 ASCII Table

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